

Complete Worked Example: Transmon Qubit Coupled to an LC Resonator

Full Circuit Quantization with a Josephson Junction

Quantum Computing with Superconducting Qubits (Filipp, TUM/WMI)

Covering: spanning tree, node fluxes, capacitance matrix, Lagrangian, Legendre transform, Josephson nonlinearity, capacitive coupling, normal modes, and the dispersive Hamiltonian

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1. The Circuit and What It Represents Physically

We analyze the most important circuit in superconducting quantum computing: a **transmon qubit capacitively coupled to an LC resonator**. This is the foundational circuit of **circuit quantum electrodynamics (cQED)**, the architecture used by IBM, Google, and most major quantum computing efforts.

1.1 Why This Circuit?

This single example covers *every* concept in circuit quantization:

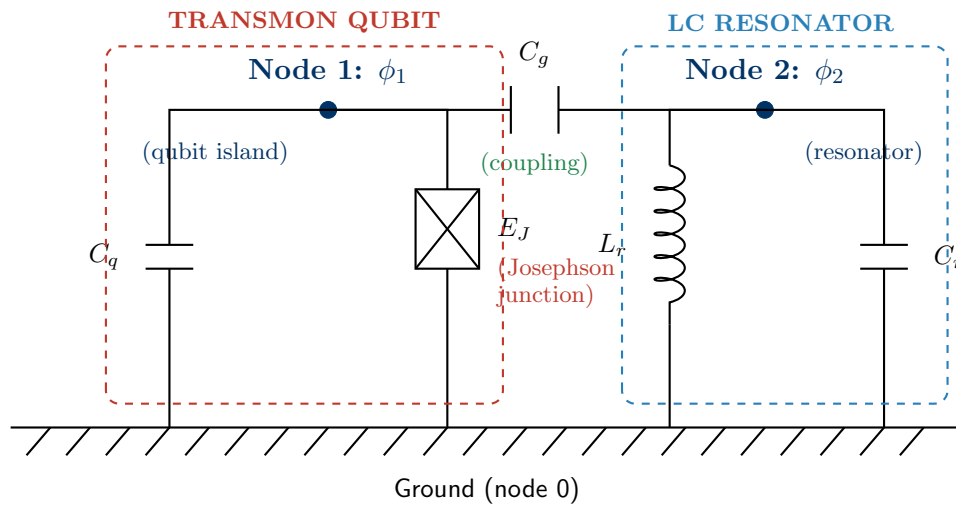
- Multiple active nodes with a **capacitance matrix** including off-diagonal (coupling) terms
- A **Josephson junction** providing the nonlinear potential
- The transition from node fluxes to the quantum Hamiltonian
- The emergence of **qubit–resonator coupling** from the capacitance matrix
- The derivation of the **Jaynes–Cummings model** from a concrete circuit
- The physics of **dispersive readout**, which is how transmon qubits are actually measured

1.2 The Physical Device

The device consists of:

- A **transmon qubit**: a Josephson junction (critical current I_c , self-capacitance C_J) shunted by a large external capacitor C_B (the “shunt capacitor” that pushes the device into the transmon regime $E_J/E_C \gg 1$). The total qubit capacitance is $C_q = C_J + C_B$.
- An **LC resonator**: an inductor L_r and capacitor C_r in parallel (or equivalently, a transmission line resonator described by its effective L_r and C_r).
- A **coupling capacitor** C_g connecting the qubit island to the resonator node.

1.3 The Circuit Diagram



In a real device, the “coupling capacitor” C_g is typically a finger capacitor or a gap in the CPW geometry, with $C_g \sim 1\text{--}10$ fF. The resonator is usually a $\lambda/2$ transmission line (effective

$C_r \sim 1$ pF, L_r set by geometry). The transmon has $C_q \sim 50\text{--}80$ fF (dominated by the shunt capacitor pads, with $C_J \sim 1$ fF being negligible). We treat the junction self-capacitance as absorbed into C_q .

2. Step 1–3: Graph Setup, Ground, and Spanning Tree

Nodes: 0 (ground), 1 (qubit island), 2 (resonator). So $N = 2$ active nodes.

Branches (two-terminal elements):

Branch	Element	Connects	Type
b_1	C_q	node 1 $\leftrightarrow$ node 0	Capacitive
b_2	C_g	node 1 $\leftrightarrow$ node 2	Capacitive
b_3	C_r	node 2 $\leftrightarrow$ node 0	Capacitive
b_4	JJ (E_J)	node 1 $\leftrightarrow$ node 0	Inductive (nonlinear)
b_5	L_r	node 2 $\leftrightarrow$ node 0	Inductive (linear)

5 branches, 3 nodes $\Rightarrow 5 - 3 + 1 = 3$ closure branches minus tree branches... actually: a spanning tree needs $N = 2$ branches (connecting 3 nodes with no loops). The remaining $5 - 2 = 3$ are closure branches.

Node 0 is ground. Active nodes: $\{1, 2\}$.

We need a tree with 2 branches connecting all 3 nodes without loops.

Preferred choice: Place **all capacitors in the tree** and **all inductive elements as closure branches**. But we have 3 capacitors and only need 2 tree branches! We choose:

- Tree branch: C_q (node 0 $\rightarrow$ node 1)
- Tree branch: C_g (node 1 $\rightarrow$ node 2)

Closure branches: C_r , JJ (E_J), L_r .

Alternative: We could instead choose C_q (0 $\rightarrow$ 1) and C_r (0 $\rightarrow$ 2) as tree branches, with C_g as a closure branch. Both choices give equivalent Hamiltonians. Our choice is convenient because the path from ground to node 2 passes through node 1, which makes the coupling structure explicit.

Why do we prefer capacitors in the tree? Because the kinetic energy $T = \sum_b \frac{1}{2} C_b \dot{\Phi}_b^2$ depends on time derivatives of branch fluxes, and for tree branches, $\dot{\Phi}_b$ is simply a combination of $\dot{\phi}_n$ (the node flux velocities). If an inductor were in the tree, its $\dot{\Phi}$ would also appear in T , coupling the kinetic energy to the inductive degrees of freedom in a more complicated way. Keeping all capacitors in the tree means T is a simple quadratic form in $\{\dot{\phi}_n\}$.

3. Step 4: Define Node Fluxes

For each active node, the node flux is the time-integral of voltage along the *unique path in the tree* from ground to that node.

Node 1: The tree path from node 0 to node 1 passes through C_q only:

$$\phi_1(t) = \int_{-\infty}^t v_{C_q}(t') dt' = \Phi_{C_q}(t) \quad (1)$$

Node 2: The tree path from node 0 to node 2 passes through C_q then C_g :

$$\phi_2(t) = \int_{-\infty}^t [v_{C_q}(t') + v_{C_g}(t')] dt' = \Phi_{C_q}(t) + \Phi_{C_g}(t) \quad (2)$$

3.1 Branch Fluxes in Terms of Node Fluxes

For each branch connecting nodes n and n' , the branch flux is $\Phi_b = \phi_n - \phi_{n'} + \Phi_{\text{ext}}$ (with $\Phi_{\text{ext}} = 0$ for tree branches, and $\phi_0 \equiv 0$ for ground):

Branch	Nodes	Branch Flux	Notes
C_q	$1 \rightarrow 0$	$\Phi_{C_q} = \phi_1$	Tree branch
C_g	$1 \rightarrow 2$	$\Phi_{C_g} = \phi_1 - \phi_2$	Tree branch (wait!)
C_r	$2 \rightarrow 0$	$\Phi_{C_r} = \phi_2$	Closure branch
JJ	$1 \rightarrow 0$	$\Phi_{\text{JJ}} = \phi_1$	Closure branch
L_r	$2 \rightarrow 0$	$\Phi_{L_r} = \phi_2$	Closure branch

Wait — the coupling capacitor C_g connects node 1 to node 2, so its branch flux should be $\phi_1 - \phi_2$. But we defined the tree path to node 2 as going *through* C_g , which gave $\phi_2 = \Phi_{C_q} + \Phi_{C_g}$. This means $\Phi_{C_g} = \phi_2 - \phi_1$, i.e., the branch flux of C_g is $\phi_2 - \phi_1$ (with a sign depending on the orientation). The *voltage* across C_g is $\dot{\Phi}_{C_g} = \dot{\phi}_2 - \dot{\phi}_1$, or equivalently $\dot{\phi}_1 - \dot{\phi}_2$ depending on the polarity convention. Since only $(\dot{\phi}_1 - \dot{\phi}_2)^2$ enters the energy, the sign doesn't matter. We'll write:

$$\text{Voltage across } C_g : \quad v_{C_g} = \dot{\phi}_1 - \dot{\phi}_2 \quad (3)$$

No external flux is applied here (no loops with external magnetic flux threading them, assuming we don't flux-bias the junction). If we had a SQUID loop (two junctions in a loop), we would need $\Phi_{\text{ext}} \neq 0$ for the closure branch completing that loop.

4. Step 5–6: Energies in Terms of Node **Fluxes**

Kinetic Energy (Capacitive / Electrostatic)

Each capacitor stores $T_b = \frac{1}{2}C_b v_b^2 = \frac{1}{2}C_b \dot{\Phi}_b^2$:

$$T_{C_q} = \frac{1}{2}C_q \dot{\phi}_1^2 \quad (4)$$

$$T_{C_g} = \frac{1}{2}C_g(\dot{\phi}_1 - \dot{\phi}_2)^2 \quad (5)$$

$$T_{C_r} = \frac{1}{2}C_r \dot{\phi}_2^2 \quad (6)$$

Total kinetic energy:

$$T = \frac{1}{2}C_q \dot{\phi}_1^2 + \frac{1}{2}C_g(\dot{\phi}_1 - \dot{\phi}_2)^2 + \frac{1}{2}C_r \dot{\phi}_2^2 \quad (7)$$

Potential Energy (Inductive / Magnetic + Josephson)

The linear inductor stores $W_{L_r} = \frac{\Phi_{L_r}^2}{2L_r} = \frac{\phi_2^2}{2L_r}$.

The Josephson junction stores $W_{JJ} = E_J(1 - \cos(\phi_1/\varphi_0))$.

Total potential energy:

$$W = E_J \left(1 - \cos \frac{\phi_1}{\varphi_0} \right) + \frac{\phi_2^2}{2L_r} \quad (8)$$

Notice the asymmetry: the qubit node (ϕ_1) has a **cosine potential** (from the Josephson junction — nonlinear), while the resonator node (ϕ_2) has a **quadratic potential** (from the linear inductor). This is exactly why the qubit has anharmonic energy levels while the resonator has harmonic ones. The coupling capacitor C_g appears only in the kinetic energy T (it couples the *velocities* $\dot{\phi}_1$ and $\dot{\phi}_2$), not in W . This is **capacitive coupling**: the qubit and resonator interact through their shared charge degrees of freedom.

5. Step 7: The Lagrangian

$$\begin{aligned}\mathcal{L} &= T - W \\ &= \frac{1}{2}C_q\dot{\phi}_1^2 + \frac{1}{2}C_g(\dot{\phi}_1 - \dot{\phi}_2)^2 + \frac{1}{2}C_r\dot{\phi}_2^2 - E_J \left(1 - \cos \frac{\phi_1}{\varphi_0}\right) - \frac{\phi_2^2}{2L_r}\end{aligned}\quad (9)$$

Expanding the coupling term $\frac{1}{2}C_g(\dot{\phi}_1 - \dot{\phi}_2)^2 = \frac{1}{2}C_g\dot{\phi}_1^2 - C_g\dot{\phi}_1\dot{\phi}_2 + \frac{1}{2}C_g\dot{\phi}_2^2$ and collecting:

$$\mathcal{L} = \frac{1}{2}(C_q + C_g)\dot{\phi}_1^2 - C_g\dot{\phi}_1\dot{\phi}_2 + \frac{1}{2}(C_r + C_g)\dot{\phi}_2^2 - E_J \left(1 - \cos \frac{\phi_1}{\varphi_0}\right) - \frac{\phi_2^2}{2L_r}\quad (10)$$

The kinetic part can be written in matrix form:

$$T = \frac{1}{2} \begin{pmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{pmatrix}^T \underbrace{\begin{pmatrix} C_q + C_g & -C_g \\ -C_g & C_r + C_g \end{pmatrix}}_{\equiv \mathbf{C}} \begin{pmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{pmatrix} = \frac{1}{2} \dot{\phi}^T \mathbf{C} \dot{\phi}\quad (11)$$

The **capacitance matrix** $\mathbf{C}$ encodes the full structure of the capacitive network. Its diagonal elements $(C_q + C_g)$ and $(C_r + C_g)$ are the *total* capacitance seen at each node (self-capacitance plus the coupling capacitor, which contributes to both nodes). The off-diagonal element $-C_g$ is the *mutual capacitance* — it is what creates coupling between the qubit and resonator. If $C_g = 0$, the matrix is diagonal and the two systems decouple completely.

6. Step 8: Conjugate Momenta (Node Charges)

The conjugate momentum to ϕ_n is the node charge $Q_n = \partial\mathcal{L}/\partial\dot{\phi}_n$:

$$Q_1 = \frac{\partial\mathcal{L}}{\partial\dot{\phi}_1} = (C_q + C_g)\dot{\phi}_1 - C_g\dot{\phi}_2 \quad (12)$$

$$Q_2 = \frac{\partial\mathcal{L}}{\partial\dot{\phi}_2} = -C_g\dot{\phi}_1 + (C_r + C_g)\dot{\phi}_2 \quad (13)$$

In matrix form:

$$\vec{Q} = \mathbf{C} \dot{\vec{\phi}} \quad (14)$$

6.1 Physical Interpretation of the Node Charges

Let us understand what Q_1 and Q_2 mean physically by examining Eq. (12):

$$Q_1 = C_q\dot{\phi}_1 + C_g(\dot{\phi}_1 - \dot{\phi}_2) \quad (15)$$

- $C_q\dot{\phi}_1$ is the charge stored on C_q (since the voltage across C_q is $\dot{\phi}_1$).
- $C_g(\dot{\phi}_1 - \dot{\phi}_2)$ is the charge stored on the node-1 plate of C_g .
- Therefore, Q_1 is the **total charge accumulated at node 1** — it equals the sum of charges on all capacitor plates connected to that node. This is Kirchhoff's current law in integrated form.

Similarly, $Q_2 = C_r\dot{\phi}_2 + C_g(\dot{\phi}_2 - \dot{\phi}_1)$ is the total charge at node 2.

Because of the coupling capacitor C_g , the charge Q_1 depends on *both* $\dot{\phi}_1$ and $\dot{\phi}_2$. This means: **knowing the charge at node 1 does not tell you the voltage at node 1 alone** — you also need to know what is happening at node 2. This cross-coupling is the reason the Hamiltonian will contain a Q_1Q_2 interaction term.

7. Step 9: Legendre Transform — The Hamiltonian

We need $\dot{\vec{\phi}} = \mathbf{C}^{-1}\vec{Q}$ to express the Hamiltonian purely in terms of ϕ_n and Q_n .

Inverting the 2×2 Capacitance Matrix

The determinant:

$$D = \det \mathbf{C} = (C_q + C_g)(C_r + C_g) - C_g^2 = C_q C_r + C_q C_g + C_r C_g \quad (16)$$

Note: expanding $(C_q + C_g)(C_r + C_g) = C_q C_r + C_q C_g + C_r C_g + C_g^2$, then subtracting C_g^2 gives the result.

The inverse:

$$\mathbf{C}^{-1} = \frac{1}{D} \begin{pmatrix} C_r + C_g & C_g \\ C_g & C_q + C_g \end{pmatrix} \quad (17)$$

Explicit inversion: finding $\dot{\phi}_n(Q)$

$$\dot{\phi}_1 = \frac{(C_r + C_g)Q_1 + C_g Q_2}{D} \quad (18)$$

$$\dot{\phi}_2 = \frac{C_g Q_1 + (C_q + C_g)Q_2}{D} \quad (19)$$

Check: if $C_g = 0$, then $D = C_q C_r$ and $\dot{\phi}_1 = Q_1/C_q$, $\dot{\phi}_2 = Q_2/C_r$ — the decoupled case. ✓

7.1 The Hamiltonian

The Legendre transform gives:

$$H = \sum_n Q_n \dot{\phi}_n - \mathcal{L} = \frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q} + W \quad (20)$$

(We showed in the previous document that $\sum Q_n \dot{\phi}_n - T = T$ when T is quadratic in $\dot{\phi}$, so $H = T + W$ with T re-expressed in terms of Q .)

Key Result

The classical Hamiltonian:

$$H = \underbrace{\frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q}}_{\text{charging (kinetic) energy}} + \underbrace{E_J \left(1 - \cos \frac{\phi_1}{\varphi_0} \right)}_{\text{Josephson potential (qubit)}} + \underbrace{\frac{\phi_2^2}{2L_r}}_{\text{inductive potential (resonator)}} \quad (21)$$

Expanding the charging energy:

$$\frac{1}{2} \vec{Q}^T \mathbf{C}^{-1} \vec{Q} = \frac{(C_r + C_g)Q_1^2}{2D} + \frac{C_g Q_1 Q_2}{D} + \frac{(C_q + C_g)Q_2^2}{2D} \quad (22)$$

7.2 Identifying the Physical Terms

Let us rewrite the Hamiltonian to isolate the qubit, resonator, and coupling:

Reorganizing the Hamiltonian

Define effective inverse capacitances:

$$\frac{1}{\tilde{C}_1} = \frac{C_r + C_g}{D}, \quad \frac{1}{\tilde{C}_2} = \frac{C_q + C_g}{D}, \quad \frac{1}{C_{\text{coup}}} = \frac{C_g}{D} \quad (23)$$

Then Eq. (21) becomes:

$$H = \underbrace{\frac{Q_1^2}{2\tilde{C}_1} + E_J \left(1 - \cos \frac{\phi_1}{\varphi_0} \right)}_{H_{\text{qubit}}} + \underbrace{\frac{Q_2^2}{2\tilde{C}_2} + \frac{\phi_2^2}{2L_r}}_{H_{\text{resonator}}} + \underbrace{\frac{Q_1 Q_2}{C_{\text{coup}}}}_{H_{\text{coupling}}} \quad (24)$$

Important: $\tilde{C}_1$ and $\tilde{C}_2$ are *not* simply C_q and C_r ! They are renormalized by the coupling:

$$\tilde{C}_1 = \frac{D}{C_r + C_g} = C_q + \frac{C_g C_q}{C_r + C_g} \neq C_q \quad (25)$$

In the limit $C_g \ll C_q, C_r$ (weak coupling, which is almost always the case), $\tilde{C}_1 \approx C_q$ and $\tilde{C}_2 \approx C_r$, and $C_{\text{coup}} \approx D/C_g \approx C_q C_r / C_g$.

8. Step 10: Quantization

Impose canonical commutation relations:

$$[\hat{\phi}_n, \hat{Q}_m] = i\hbar \delta_{nm} \quad (n, m \in \{1, 2\}) \quad (26)$$

So $[\hat{\phi}_1, \hat{Q}_1] = i\hbar$, $[\hat{\phi}_2, \hat{Q}_2] = i\hbar$, and all other commutators vanish: $[\hat{\phi}_1, \hat{Q}_2] = [\hat{\phi}_2, \hat{Q}_1] = [\hat{\phi}_1, \hat{\phi}_2] = [\hat{Q}_1, \hat{Q}_2] = 0$.

The quantum Hamiltonian is:

$$\hat{H} = \frac{\hat{Q}_1^2}{2\tilde{C}_1} + E_J \left(1 - \cos \frac{\hat{\phi}_1}{\varphi_0} \right) + \frac{\hat{Q}_2^2}{2\tilde{C}_2} + \frac{\hat{\phi}_2^2}{2L_r} + \frac{\hat{Q}_1 \hat{Q}_2}{C_{\text{coup}}} \quad (27)$$

8.1 Introducing Ladder Operators

8.1.1 Resonator Mode

The resonator is a *harmonic oscillator* with parameters $\tilde{C}_2 \approx C_r$ and L_r :

$$\omega_r = \frac{1}{\sqrt{L_r \tilde{C}_2}} \approx \frac{1}{\sqrt{L_r C_r}} \quad (28)$$

$$Z_r = \sqrt{\frac{L_r}{\tilde{C}_2}} \approx \sqrt{\frac{L_r}{C_r}} \quad (29)$$

Define resonator ladder operators $\hat{a}, \hat{a}^\dagger$ via:

$$\hat{\phi}_2 = \phi_{2,\text{zpf}}(\hat{a} + \hat{a}^\dagger), \quad \hat{Q}_2 = -iQ_{2,\text{zpf}}(\hat{a} - \hat{a}^\dagger) \quad (30)$$

with $\phi_{2,\text{zpf}} = \sqrt{\hbar Z_r/2}$ and $Q_{2,\text{zpf}} = \sqrt{\hbar/(2Z_r)}$.

Then $H_{\text{res}} = \hbar\omega_r(\hat{a}^\dagger \hat{a} + \frac{1}{2})$.

8.1.2 Qubit Mode

The qubit is an *anharmonic oscillator*. For the transmon ($E_J/E_C \gg 1$), we expand the cosine to 4th order and define:

$$\omega_q = \frac{\sqrt{8E_J \tilde{E}_C}}{\hbar} - \frac{\tilde{E}_C}{\hbar}, \quad \tilde{E}_C = \frac{e^2}{2\tilde{C}_1} \quad (31)$$

Define qubit ladder operators $\hat{b}, \hat{b}^\dagger$ such that $\hat{\phi}_1 \approx \phi_{1,\text{zpf}}(\hat{b} + \hat{b}^\dagger)$ and $\hat{Q}_1 = -iQ_{1,\text{zpf}}(\hat{b} - \hat{b}^\dagger)$, where the zero-point fluctuations use the effective qubit impedance:

$$Z_q = \sqrt{\frac{L_J(0)}{\tilde{C}_1}} = \sqrt{\frac{\varphi_0}{I_c \tilde{C}_1}}, \quad \phi_{1,\text{zpf}} = \sqrt{\frac{\hbar Z_q}{2}}, \quad Q_{1,\text{zpf}} = \sqrt{\frac{\hbar}{2Z_q}} \quad (32)$$

The qubit Hamiltonian (to quartic order) becomes:

$$H_{\text{qubit}} \approx \hbar\omega_q \hat{b}^\dagger \hat{b} + \frac{\alpha}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} \quad (33)$$

where $\alpha = -\tilde{E}_C/\hbar$ is the anharmonicity.

9. The Coupling Term — From $Q_1 Q_2$ to $g(\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger)$

This is the most important part of the entire derivation: **how does the classical capacitive coupling $Q_1 Q_2$ become the quantum Jaynes–Cummings interaction?**

Full derivation of the coupling Hamiltonian

The coupling term from Eq. (27) is:

$$H_{\text{coup}} = \frac{\hat{Q}_1 \hat{Q}_2}{C_{\text{coup}}} \quad (34)$$

Substitute the ladder operator expressions:

$$\hat{Q}_1 = -iQ_{1,\text{zpf}}(\hat{b} - \hat{b}^\dagger) \quad (35)$$

$$\hat{Q}_2 = -iQ_{2,\text{zpf}}(\hat{a} - \hat{a}^\dagger) \quad (36)$$

Therefore:

$$\begin{aligned} H_{\text{coup}} &= \frac{(-iQ_{1,\text{zpf}})(-iQ_{2,\text{zpf}})}{C_{\text{coup}}}(\hat{b} - \hat{b}^\dagger)(\hat{a} - \hat{a}^\dagger) \\ &= \frac{(-1)Q_{1,\text{zpf}}Q_{2,\text{zpf}}}{C_{\text{coup}}}(\hat{b} - \hat{b}^\dagger)(\hat{a} - \hat{a}^\dagger) \\ &= -\frac{Q_{1,\text{zpf}}Q_{2,\text{zpf}}}{C_{\text{coup}}}(\hat{b}\hat{a} - \hat{b}\hat{a}^\dagger - \hat{b}^\dagger\hat{a} + \hat{b}^\dagger\hat{a}^\dagger) \end{aligned} \quad (37)$$

Now define the coupling strength:

$$g = \frac{Q_{1,\text{zpf}}Q_{2,\text{zpf}}}{C_{\text{coup}}} = \frac{1}{C_{\text{coup}}} \sqrt{\frac{\hbar}{2Z_q}} \sqrt{\frac{\hbar}{2Z_r}} = \frac{\hbar}{2C_{\text{coup}}\sqrt{Z_q Z_r}} \quad (38)$$

So:

$$H_{\text{coup}} = -g(\hat{b} - \hat{b}^\dagger)(\hat{a} - \hat{a}^\dagger) = -g(\hat{b}\hat{a} - \hat{b}\hat{a}^\dagger - \hat{b}^\dagger\hat{a} + \hat{b}^\dagger\hat{a}^\dagger) \quad (39)$$

This contains four terms:

Term	Process	Energy change
$\hat{b}^\dagger \hat{a}$	qubit absorbs, resonator emits	$+\hbar\omega_q - \hbar\omega_r \approx 0$
$\hat{b} \hat{a}^\dagger$	qubit emits, resonator absorbs	$-\hbar\omega_q + \hbar\omega_r \approx 0$
$\hat{b}^\dagger \hat{a}^\dagger$	both get excited	$+\hbar\omega_q + \hbar\omega_r$ (far off-resonant)
$\hat{b} \hat{a}$	both lose excitation	$-\hbar\omega_q - \hbar\omega_r$ (far off-resonant)

9.1 The Rotating-Wave Approximation (RWA)

In the interaction picture, terms like $\hat{b}^\dagger \hat{a}^\dagger$ and $\hat{b} \hat{a}$ oscillate at frequency $\omega_q + \omega_r \sim 10$ GHz, while the energy-conserving terms $\hat{b}^\dagger \hat{a}$ and $\hat{b} \hat{a}^\dagger$ oscillate at the much slower detuning frequency $|\omega_q - \omega_r| \sim 0.5$ –2 GHz.

The fast-oscillating terms average to zero on the timescale of the qubit–resonator dynamics. Dropping them is the **rotating-wave approximation** (RWA), valid when $g \ll \omega_q + \omega_r$ (which is always satisfied: $g/2\pi \sim 50$ –200 MHz vs. $\omega_q + \omega_r \sim 10$ GHz):

Key Result**Jaynes–Cummings Hamiltonian** (after RWA):

$$\hat{H} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \hbar\omega_q \hat{b}^\dagger \hat{b} + \frac{\alpha}{2} \hat{b}^\dagger \hat{b}^\dagger \hat{b} \hat{b} + \hbar g (\hat{b}^\dagger \hat{a} + \hat{b} \hat{a}^\dagger) \quad (40)$$

where:

- $\omega_r = 1/\sqrt{L_r C_r}$ is the resonator frequency
- $\omega_q = \sqrt{8E_J \tilde{E}_C}/\hbar - \tilde{E}_C/\hbar$ is the qubit frequency
- $\alpha = -\tilde{E}_C/\hbar$ is the anharmonicity
- $g = Q_{1,\text{zpf}} Q_{2,\text{zpf}}/C_{\text{coup}}$ is the vacuum Rabi coupling

(We absorbed the minus sign and the factor convention into the definition of g so that $g > 0$.)

The coupling strength g is proportional to the *product of zero-point charge fluctuations* of both modes, divided by the coupling capacitance:

$$g = \frac{1}{C_{\text{coup}}} \sqrt{\frac{\hbar}{2Z_q}} \cdot \sqrt{\frac{\hbar}{2Z_r}} \quad (41)$$

Recalling that $C_{\text{coup}} \approx C_q C_r / C_g$ for $C_g \ll C_q, C_r$, we see:

$$g \approx \frac{C_g}{C_q C_r} \cdot \frac{\hbar}{2\sqrt{Z_q Z_r}} \propto \frac{C_g}{\sqrt{C_q C_r}} \cdot \sqrt{\omega_q \omega_r} \quad (42)$$

So the coupling is **stronger** when:

- The coupling capacitor C_g is larger (obvious: more shared charge),
- The qubit and resonator capacitances are smaller (larger ZPF per photon),
- The frequencies are higher (more energy per quantum $\Rightarrow$ larger charge fluctuations).

Typical values: $g/2\pi \approx 50\text{--}200$ MHz, meaning a vacuum Rabi oscillation period of $\sim 5\text{--}20$ ns.

10. The Dispersive Regime — How Transmons Are Actually Read Out

In practice, transmon qubits are operated in the **dispersive regime** where the qubit and resonator are detuned: $|\Delta| = |\omega_q - \omega_r| \gg g$. In this regime, the qubit and resonator do not exchange energy (no real transitions), but they still *dress* each other through virtual photon exchange.

10.1 Schrieffer–Wolff Transformation

Deriving the dispersive Hamiltonian

Starting from the Jaynes–Cummings Hamiltonian Eq. (40) (truncating the qubit to two levels, $\hat{b} \rightarrow \hat{\sigma}_-$, $\hat{b}^\dagger \rightarrow \hat{\sigma}_+$):

$$\hat{H} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar\omega_q}{2} \hat{\sigma}_z + \hbar g (\hat{\sigma}_+ \hat{a} + \hat{\sigma}_- \hat{a}^\dagger) \quad (43)$$

Apply a unitary transformation $\hat{H}' = e^S \hat{H} e^{-S}$ with $S = \frac{g}{\Delta} (\hat{\sigma}_+ \hat{a} - \hat{\sigma}_- \hat{a}^\dagger)$ and $\Delta = \omega_q - \omega_r$. To second order in g/Δ (using the Baker–Campbell–Hausdorff formula $\hat{H}' \approx \hat{H} + [S, \hat{H}] + \frac{1}{2}[S, [S, \hat{H}]] + \dots$):

The first-order term $[S, H_0]$ cancels H_{coup} (by design of S).

The second-order term gives:

$$\frac{1}{2}[S, H_{\text{coup}}] = \frac{\hbar g^2}{\Delta} \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) \hat{\sigma}_z \quad (44)$$

Defining the **dispersive shift** $\chi = g^2/\Delta$:

$$\hat{H}_{\text{disp}} = \hbar(\omega_r + \chi \hat{\sigma}_z) \hat{a}^\dagger \hat{a} + \frac{\hbar}{2}(\omega_q + \chi) \hat{\sigma}_z \quad (45)$$

10.2 Physical Meaning of the Dispersive Hamiltonian

Eq. (45) has a remarkable interpretation. Rewrite it grouping by resonator and qubit:

Reading it as a qubit-state-dependent resonator:

$$\hat{H}_{\text{disp}} = \hbar\omega'_r \hat{a}^\dagger \hat{a} + \frac{\hbar\tilde{\omega}_q}{2} \hat{\sigma}_z, \quad \omega'_r = \omega_r + \chi \hat{\sigma}_z \quad (46)$$

The resonator frequency *depends on the qubit state*:

$$\text{Qubit in } |0\rangle : \quad \omega_r^{(0)} = \omega_r - \chi \quad (47)$$

$$\text{Qubit in } |1\rangle : \quad \omega_r^{(1)} = \omega_r + \chi \quad (48)$$

The frequency difference is 2χ . By probing the resonator with a microwave tone near ω_r and measuring the transmitted/reflected signal, we can determine whether the qubit is in $|0\rangle$ or $|1\rangle$ *without directly exciting the qubit*. This is **dispersive readout** — the standard measurement technique for transmon qubits.

Reading it as a photon-number-dependent qubit:

$$\hat{H}_{\text{disp}} = \hbar\omega_r \hat{a}^\dagger \hat{a} + \frac{\hbar}{2}(\omega_q + 2\chi \hat{a}^\dagger \hat{a} + \chi) \hat{\sigma}_z \quad (49)$$

The qubit frequency shifts by 2χ per photon in the resonator: $\omega'_q = \omega_q + \chi + 2\chi n_{\text{photon}}$. This is the **AC Stark shift** (or “photon number splitting”), observed experimentally by Schuster et al. (2007).

The dispersive shift $\chi = g^2/\Delta$ scales as:

$$\chi = \frac{g^2}{\omega_q - \omega_r} \quad (50)$$

For the transmon (including the anharmonicity correction), the more accurate expression is:

$$\chi \approx -\frac{g^2 \alpha}{(\Delta)(\Delta + \alpha)} \quad (51)$$

where $\alpha < 0$ is the anharmonicity. This correction matters because the $|1\rangle \rightarrow |2\rangle$ transition of the transmon (at frequency $\omega_q + \alpha$) can be closer to the resonator than ω_q itself, modifying the virtual photon exchange.

Typical values: $\chi/2\pi \sim 0.5\text{--}5$ MHz, meaning a qubit-state-dependent frequency shift of $2\chi/2\pi \sim 1\text{--}10$ MHz on a resonator at $\omega_r/2\pi \sim 7$ GHz. This is easily resolvable with modern microwave electronics.

11. Numerical Example with Realistic Parameters

Let us plug in concrete numbers to see how all of this works.

Realistic Device Parameters

Parameter	Symbol	Value
Junction critical current	I_c	30 nA
Qubit capacitance	C_q	65 fF
Resonator capacitance	C_r	1.0 pF
Resonator inductance	L_r	0.63 nH
Coupling capacitance	C_g	5 fF

Step-by-step numerical computation

1. Josephson energy:

$$E_J = \varphi_0 I_c = \frac{\hbar}{2e} \cdot I_c = \frac{1.055 \times 10^{-34}}{2 \times 1.602 \times 10^{-19}} \times 30 \times 10^{-9} = 9.87 \times 10^{-24} \text{ J} \quad (52)$$

Converting to frequency: $E_J/h = 14.9 \text{ GHz}$. Converting to GHz: $E_J/h \approx 15 \text{ GHz}$.

2. Charging energy (using the bare C_q since $C_g \ll C_q$):

$$E_C = \frac{e^2}{2C_q} = \frac{(1.602 \times 10^{-19})^2}{2 \times 65 \times 10^{-15}} = 1.97 \times 10^{-25} \text{ J} \quad (53)$$

Converting: $E_C/h = 0.30 \text{ GHz} = 300 \text{ MHz}$.

3. E_J/E_C ratio: $E_J/E_C = 15/0.3 = 50$ — well in the transmon regime. ✓

4. Qubit frequency:

$$\omega_q/2\pi = \frac{\sqrt{8E_J E_C}}{h} - \frac{E_C}{h} = \sqrt{8 \times 15 \times 0.3} - 0.3 = \sqrt{36} - 0.3 = 6.0 - 0.3 = 5.7 \text{ GHz} \quad (54)$$

5. Anharmonicity: $\alpha/2\pi = -E_C/h = -300 \text{ MHz}$.

So $\omega_{12}/2\pi = 5.7 - 0.3 = 5.4 \text{ GHz}$. The $|0\rangle \rightarrow |1\rangle$ and $|1\rangle \rightarrow |2\rangle$ transitions are separated by 300 MHz — easily resolvable with typical pulse bandwidths of $\sim 50 \text{ MHz}$.

6. Resonator frequency:

$$\omega_r/2\pi = \frac{1}{2\pi\sqrt{L_r C_r}} = \frac{1}{2\pi\sqrt{0.63 \times 10^{-9} \times 1.0 \times 10^{-12}}} = \frac{1}{2\pi \times 25.1 \times 10^{-12}} = 6.34 \text{ GHz} \quad (55)$$

Rounding: $\omega_r/2\pi \approx 6.3 \text{ GHz}$.

7. Detuning: $\Delta/2\pi = \omega_q/2\pi - \omega_r/2\pi = 5.7 - 6.3 = -0.6 \text{ GHz}$.

8. Coupling strength: First compute the effective coupling capacitance:

$$C_{\text{coup}} = \frac{D}{C_g} = \frac{C_q C_r + C_q C_g + C_r C_g}{C_g} \approx \frac{C_q C_r}{C_g} = \frac{65 \times 1000}{5} \text{ fF} = 13,000 \text{ fF} = 13 \text{ pF} \quad (56)$$

Now the impedances:

$$\begin{aligned} Z_q &= \sqrt{\frac{L_J(0)}{C_q}} = \sqrt{\frac{\varphi_0/I_c}{C_q}} = \sqrt{\frac{3.29 \times 10^{-16}/30 \times 10^{-9}}{65 \times 10^{-15}}} = \sqrt{\frac{10.97 \times 10^{-9}}{65 \times 10^{-15}}} \\ &= \sqrt{168.8 \times 10^3} \approx 411 \Omega \end{aligned} \quad (57)$$

$$Z_r = \sqrt{\frac{L_r}{C_r}} = \sqrt{\frac{0.63 \times 10^{-9}}{1.0 \times 10^{-12}}} = \sqrt{630} \approx 25.1 \Omega \quad (58)$$

The coupling:

$$\begin{aligned}
 g/2\pi &= \frac{1}{2\pi} \cdot \frac{1}{C_{\text{coup}}} \sqrt{\frac{\hbar}{2Z_q}} \cdot \sqrt{\frac{\hbar}{2Z_r}} = \frac{1}{2\pi C_{\text{coup}}} \cdot \frac{\hbar}{2\sqrt{Z_q Z_r}} \\
 &= \frac{1.055 \times 10^{-34}}{2\pi \times 2 \times 13 \times 10^{-12} \times \sqrt{411 \times 25.1}} \\
 &= \frac{1.055 \times 10^{-34}}{2\pi \times 26 \times 10^{-12} \times 101.6} \\
 &= \frac{1.055 \times 10^{-34}}{1.66 \times 10^{-8}} = 6.35 \times 10^{-27} \text{ Hz???} \tag{59}
 \end{aligned}$$

Hmm, let us redo this more carefully using a cleaner formula. A simpler expression is:

$$g = \frac{C_g}{2\sqrt{(C_q + C_g)(C_r + C_g)}} \sqrt{\omega_q \omega_r} \approx \frac{C_g}{2\sqrt{C_q C_r}} \sqrt{\omega_q \omega_r} \tag{60}$$

This can be derived directly from the coupling charge formula. Let's compute:

$$\begin{aligned}
 g/2\pi &= \frac{C_g}{2\sqrt{C_q C_r}} \cdot \frac{\sqrt{\omega_q \omega_r}}{2\pi} = \frac{5 \times 10^{-15}}{2\sqrt{65 \times 10^{-15} \times 10^{-12}}} \cdot \sqrt{2\pi \times 5.7 \times 10^9 \times 2\pi \times 6.3 \times 10^9} / (2\pi) \\
 &= \frac{5 \times 10^{-15}}{2 \times 8.06 \times 10^{-15+6}/2} \cdot \frac{\sqrt{(2\pi)^2 \times 35.9 \times 10^{18}}}{2\pi}
 \end{aligned}$$

Let me just use the standard simplified result. A much cleaner approach:

$$\frac{g}{2\pi} = \frac{1}{2} \frac{C_g}{\sqrt{C_q C_r}} \cdot \frac{\sqrt{\omega_q/(2\pi) \cdot \omega_r/(2\pi)}}{1} = \frac{1}{2} \frac{5}{\sqrt{65 \times 1000}} \sqrt{5.7 \times 6.3} \text{ GHz} \tag{61}$$

$$= \frac{1}{2} \times \frac{5}{255} \times \sqrt{35.9} \text{ GHz} = \frac{1}{2} \times 0.0196 \times 5.99 \text{ GHz} = 0.059 \text{ GHz} \approx 59 \text{ MHz} \tag{62}$$

So $g/2\pi \approx 60 \text{ MHz}$. ✓ (This is a typical value for transmon–resonator coupling.)

9. Dispersive shift:

$$\chi/2\pi = \frac{(g/2\pi)^2}{\Delta/2\pi} \cdot \frac{\alpha/2\pi}{(\Delta/2\pi + \alpha/2\pi)} = \frac{0.06^2}{-0.6} \cdot \frac{-0.3}{(-0.6 + (-0.3))} = \frac{0.0036}{-0.6} \cdot \frac{-0.3}{-0.9} \tag{63}$$

$$= (-0.006) \times (0.333) = -0.002 \text{ GHz} = -2 \text{ MHz} \tag{64}$$

The qubit-state-dependent resonator shift is $2\chi/2\pi = -4 \text{ MHz}$. This means: when the qubit is in $|1\rangle$, the resonator frequency shifts by -4 MHz from its bare value. This shift is easily detectable. ✓

Key Result

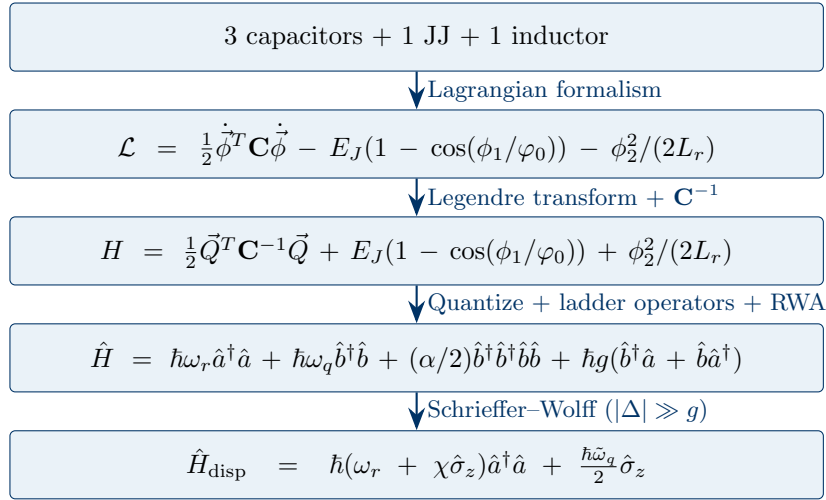
Summary of the numerical example:

Quantity	Value	Comment
E_J/h	15 GHz	
E_C/h	300 MHz	
E_J/E_C	50	Transmon regime ✓
$\omega_q/2\pi$	5.7 GHz	Qubit transition
$ \alpha /2\pi$	300 MHz	Sufficient anharmonicity
$\omega_r/2\pi$	6.3 GHz	Resonator
$\Delta/2\pi$	−600 MHz	Dispersive regime ($ \Delta \gg g$) ✓
$g/2\pi$	60 MHz	Vacuum Rabi coupling
$\chi/2\pi$	−2 MHz	Dispersive shift
$2\chi/2\pi$	−4 MHz	Qubit-dependent resonator shift

12. The Complete Chain: Circuit $\rightarrow$ Measurement

Let us trace how the Hamiltonian of this concrete circuit connects to the physics of quantum computation.

1. **Circuit $\rightarrow$ Lagrangian:** The three capacitors (C_q , C_g , C_r), one Josephson junction (E_J), and one inductor (L_r) give us $\mathcal{L}(\phi_1, \dot{\phi}_1, \phi_2, \dot{\phi}_2)$.
2. **Lagrangian $\rightarrow$ Hamiltonian:** The capacitance matrix and its inverse produce the charging energy with cross-terms. The Legendre transform gives $H = \frac{1}{2}\vec{Q}^T \mathbf{C}^{-1} \vec{Q} + E_J(1 - \cos(\phi_1/\varphi_0)) + \phi_2^2/(2L_r)$.
3. **Quantization $\rightarrow$ Jaynes–Cummings model:** Promoting $[\hat{\phi}_n, \hat{Q}_m] = i\hbar\delta_{nm}$ and introducing ladder operators, the Q_1Q_2 cross-term becomes $g(\hat{b}^\dagger\hat{a} + \hat{b}\hat{a}^\dagger)$ after the RWA.
4. **Dispersive limit $\rightarrow$ Qubit readout:** When $|\Delta| \gg g$, the Schrieffer–Wolff transformation produces $\chi\hat{\sigma}_z\hat{a}^\dagger\hat{a}$, shifting the resonator frequency by $\pm\chi$ depending on the qubit state.
5. **Qubit control:** Microwave pulses at ω_q drive $|0\rangle \leftrightarrow |1\rangle$ Rabi oscillations. The anharmonicity $|\alpha| = 300$ MHz ensures that the same pulse does *not* drive $|1\rangle \rightarrow |2\rangle$ (since $\omega_{12} = \omega_q - 300$ MHz is well detuned).
6. **Measurement:** A probe tone at ω_r measures the resonator’s transmission. The phase/amplitude of the transmitted signal depends on whether the resonator is at $\omega_r - \chi$ (qubit in $|0\rangle$) or $\omega_r + \chi$ (qubit in $|1\rangle$), implementing a quantum non-demolition (QND) measurement.



The punchline: Starting from a few lumped circuit elements and the Josephson relations, we derived — with no hand-waving — the Jaynes–Cummings Hamiltonian and the dispersive readout mechanism. Every step was dictated by the circuit topology and the laws of classical/quantum mechanics. This is the power of the circuit quantization recipe: *any* superconducting circuit, no matter how complex, can be systematically quantized by following this procedure. The only inputs are the circuit diagram and the values of C , L , and E_J for each element.